P5 – Design Space Exploration

04.12.2025, kuex

After the labs of the last weeks, it's now time to evaluate the results and decide on which AES implementation best fits our needs. We used the Ultra96 evaluation board to check implementation alternatives. To not just follow our gut feelings, we will prepare the necessary data and provide a visualization of the different solutions.

As a preparation, let's recap on the exercises with AES. We have been working on implementations of AES encryption and decryption for the following 4 scenarios:

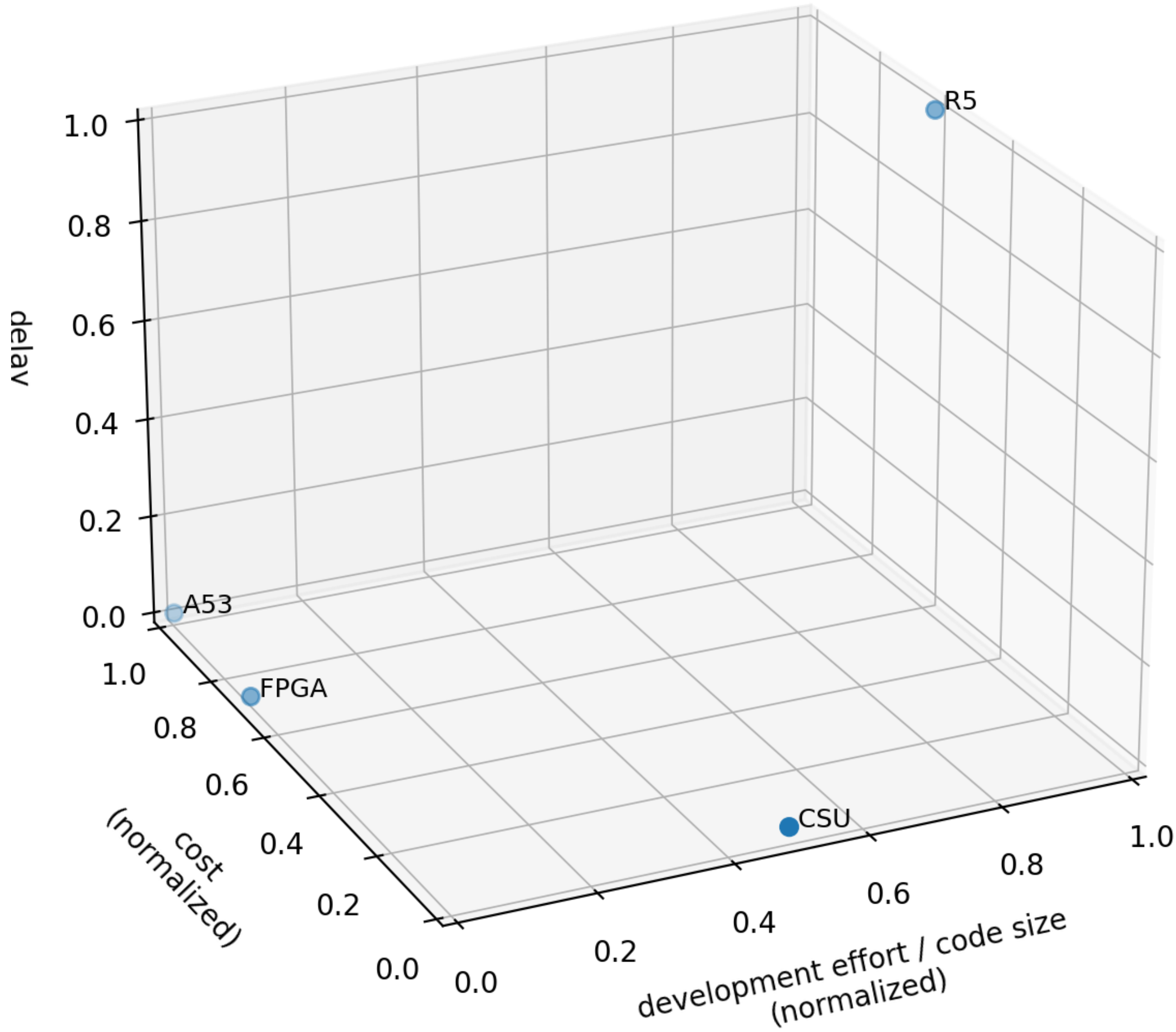
1) Implementation on a Cortex-A53
2) Implementation on a Cortex-R5
3) Implementation on an FPGA
4) Implementation on CSU

Tasks:

- Prepare a table containing timing information, code size, and cost of the different design alternatives:

Solution	Cost	Delay (runtime)	Code Size (binaries)
Cortex-A53	30 Fr. avg.	0.159 ms	75 KB
Cortex-R5	19 Fr. avg.	11.24 ms	688 KB
FPGA	22.10 Fr.	0.118 ms	74 KB
CSU	3.40 Fr.	0.275 ms	375 KB

  - Concerning cost, check unit prices at, e.g., https://digikey.ch, and use a secure element SE050 as a replacement for the CSU. For the FPGA use an XC6SLX4-2CSG225C Spartan 6
  - Concerning the delay/runtime, check your timing measurements for encryption and decryption of 64-byte packets for the different implementations. If you are missing data, re-take the measurement or talk to your colleagues.
- Normalize the values in the table such that for each dimension (cost, delay, code size) the maximum value observed is 1.
- Store the normalized values in a text file with three columns (cost, delay, code size)



delay
cost
(normalized)
development effort / code size
(normalized)
R5
A53
FPGA
CSU
0.0
0.2
0.4
0.6
0.8
1.0